

Adsorption induced magnetic anisotropy in the two-dimensional magnet CrCl₃

M. Luo^a, Y.D. Li^b, K.J. Wang^c, Y.H. Shen^{c,*}

^a Department of Physics, Shanghai Polytechnic University, Shanghai, 201209, China

^b School of Microelectronic of Fudan University, Shanghai, 200433, China

^c Key Laboratory of Polar Materials and Devices, East China Normal University, Shanghai, 200241, China

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ABSTRACT

Two-dimensional (2D) CrCl₃ has attracted much attention as it is reported to be a ferromagnetic semiconductor. By performing first-principles calculations, we investigate the effects of Li and F adsorption on the saturation magnetization and magnetic anisotropy energy (MAE) of CrCl₃. It is observed that Li adsorption can dramatically enhance its saturation magnetization, and tune its easy magnetization axis to the in-plane direction from original out-of-plane. The monotonic enhancement of in-plane magnetism in CrCl₃ as the coverage of Li increases are attributed to electrostatic doping induced by charge transfer between Li atoms and Br atoms. By contrast, the F adsorption reduces the out-of-plane magnetism in CrCl₃ as the coverage of F increases but keeps the original easy magnetization. These findings will benefit for further understanding of the tunable magnetic properties of CrCl₃ and push 2D spintronics applications.

1. Introduction

Recent discoveries of two-dimensional (2D) materials have launched great concern theoretically and experimentally because of their unique electronic and optoelectronic properties [1–5]. However, none of conventional 2D materials is reported to exhibit intrinsic magnetism, which limits their application in spintronics [6]. Large numbers of experimental and theoretical have been carried out on introducing magnetism into nonmagnetic 2D materials or seeking 2D materials with intrinsic magnetism. On the one hand, the general methods of inducing magnetism in nonmagnetic 2D materials were substitutional doping [7,8], defect engineering [9], magnetic proximity effect [10], and so on. Although some achievements have been made by these approaches, the introduced magnetism is relatively weak and difficult to control experimentally. On the other hand, a variety of theoretical prediction about new 2D ferromagnetic materials, such as CrI₃ [11,12], Cr₂Ge₂Te₆ [13], VSe₂ [14], MnSex [15], and Fe₃GeTe₂ [16], Fe₅GeTe₂ [17], have been reported. These findings provide new exciting opportunities for future spintronic applications.

Magnetic anisotropy energy (MAE) plays a key role in 2D magnetic materials, and it could overcome the thermal fluctuations and stabilizes the magnetic order at the finite temperature which has been verified by

the Mermin–Wagner theorem [18]. Moreover, ferromagnetic 2D materials with large anisotropy come to attract growing attention for their potential applications in spintronic devices. It is especially attractive to explore whether intrinsic ferromagnetism and the magnetic anisotropy of 2D monolayer can be modulated. Previous works indicated that surface adsorption is an attractive approach to control electronic structure and magnetism in 2D materials [19]. By adsorption, a charge transfer can be induced between adatoms and the neighbor atoms of host materials, leading to electrostatic doping. As is well known, Li is a typical electron donor, likely transfers charge to the substrate when adsorbed. More recently, many works have been carried out to investigate the MAE in intrinsic magnetic materials. Richter et al. observed temperature-dependent magnetic anisotropy in CrI₃ and CrBr₃ [20]. The magnetic anisotropy in monolayer CrCl₃, CrBr₃, and CrI₃ could be regulated by the strain [21]. Kim et al. studied the magnetic coupling in 2D-CrX₃ (X = I, Br, Cl) by using the combination of tunneling and magneto-optical measurements [22]. Yang et al. reported nonmetallic atoms induced magnetic anisotropy in monolayer CrBr₃ and CrI₃ [23].

As one of the intrinsic magnetic materials, the study of CrCl₃ is still limited. CrCl₃, CrBr₃, and CrI₃ have several similarities, such as FM and insulating properties. In this work, the impact of Li adsorption on the electronic and magnetic properties of monolayer CrCl₃ are investigated

* Corresponding author.

E-mail address: shenyh2016@gmail.com (Y.H. Shen).

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by first-principles calculations. We find that the easy axis of monolayer CrCl_3 can be tuned from the out-of-plane to in-plane after Li adsorption, and the MAE reaches as large as 3 times greater than that of pristine CrCl_3 . When the monolayer CrCl_3 is adsorbed by F, a typical hole contributor, the MAE also increases drastically, although the easy axis maintains out-of-plane. Our study illustrates the promising potential of electrostatic doping induced by charge transfer in tuning the magnetization orientation and enhancing magnetism in monolayer CrCl_3 .

2. Method

All calculations were carried out in the framework of density functional theory (DFT) within the generalized gradient approximation (GGA) of Perdew-Burke-Ernzerhof (PBE) [24] implemented in the Vienna ab initio simulation package (VASP) [25]. The projector-augmented wave (PAW) [26] method was used to describe the ion-electron interaction. The plane-wave cut-off energy was set to be 500 eV. The Brillouin zone was sampled by using a $10 \times 10 \times 1$ -centered Monkhorst-Pack k-point grid for the unit cell and $5 \times 5 \times 1$ k-point for $3 \times 2 \times 1$ supercell. All the atomic structures were fully relaxed until the force on each atom is smaller than 0.01 eV/Å and the total energy was converged to be within at least 10^{-6} eV (10^{-8} eV for MAE calculations). The optimized lattice constants are $a = b = 6.056$ Å, and the Cr-Cl bond length is about 2.352 Å [21]. To avoid the artificial interactions between image layers, a vacuum space of 20 Å is added along with the out of the plane z-axis. Spin-orbit coupling was taken into account to obtain the MAE. Convergence test has been carefully carried out for k-point sampling on MAE. The results show that the values of MAE converge to 10^{-5} eV/Cr as cut-off energy changes from 450 eV to 550 eV, and k mesh from $5 \times 5 \times 1$ to $10 \times 10 \times 1$ for pristine CrCl_3 .

3. Results and discussion

3.1. Magnetic properties of pristine CrCl_3

In this part, we investigated the structure and magnetism of the pristine CrCl_3 monolayer, as shown in Fig. 1(a) and (b). It consists of 2D honeycomb network of Cr atoms, which are sandwiched between two atomic planes of Cl. Moreover, Cr atoms are coordinated by edge-sharing octahedra with six Cl atoms. Our calculation shows that pristine CrCl_3 has a direct bandgap of 1.59 eV, which is consistent with the

earlier theoretical result [21]. The pristine CrCl_3 also shows large magnetism as large as about $3.0 \mu_B/\text{Cr}$. As shown in Fig. 1(c), from the projected density of states (PDOS), we can see the magnetism is dominated by the hybrid orbitals of Cr-d and Cl-p.

As is well known, MAE is an important parameter to characterize FM materials besides the magnetization and TC. In the previous work, it has been found that the magnetic properties of CrI_3 monolayer are sensitive to charge doping [27]. Inspired by this, we deem that whether the MAE of CrCl_3 could be regulated by charge doping. We calculated the MAE with charge doping by only adding electrons or holes to pristine CrCl_3 (see Fig. 2). Here, the MAE is evaluated based on the total energy difference between in-plane [100] and out-of-plane [001] magnetization direction, named $\text{MAE} = E[100] - E[001]$. Thus, the positive and negative values of MAE denote that the easy magnetization axis is perpendicular and parallel to the plane of monolayer CrCl_3 , respectively. For the pristine CrCl_3 , the calculated MAE is about 0.789 meV/Cr, the positive value indicates an out-of-plane easy magnetization axis. The MAE can be largely tuned by charge doping. The magnetization direction transforms

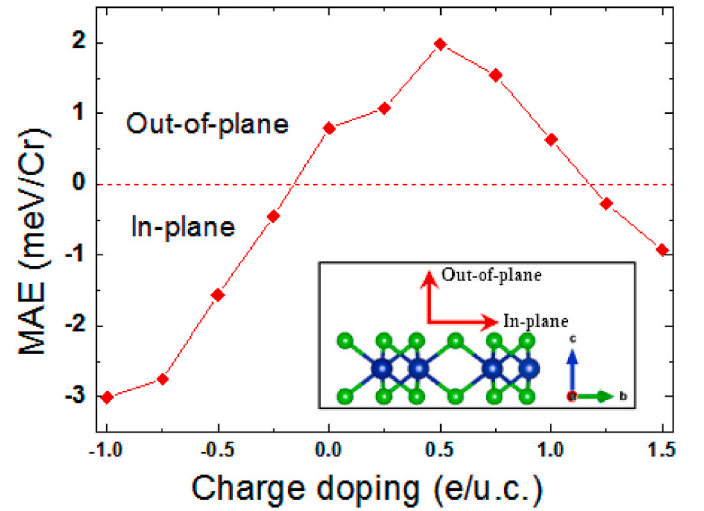


Fig. 2. The MAE of monolayer CrCl_3 with different Charge doping. The zero doping indicates the pristine system without charge doping. The negative and positive charge doping represent electron and hole doping, respectively.

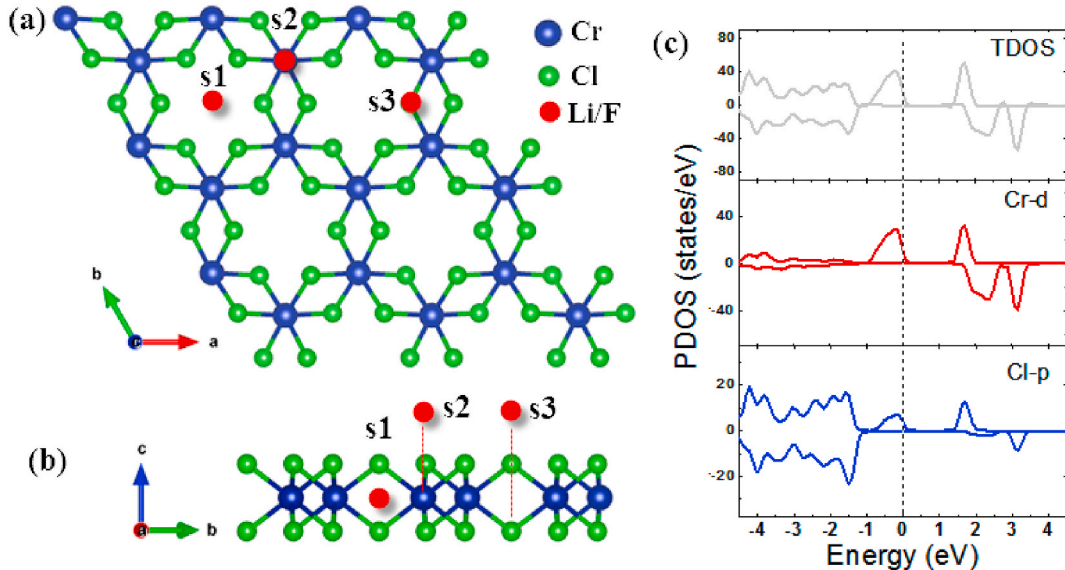


Fig. 1. (a) Top and (b) side views of pristine CrCl_3 monolayer and three possible adsorption sties of Li are marked. s1) hollow, s2) Cr-top, s3) Cl-top. (c) Projected density of states (PDOS) for CrCl_3 monolayer. The Fermi level is set to zero as marked with dot lines.

from the out-of-plane to the in-plane easy after a certain electron doping (~ -0.056 e/u.c.). While for the hole doping, the easy axis changes from out-of-plane to in-plane when a certain doping concentration is larger than a certain value (~ 1.224 e/u.c.). After studying the hypothetical doping monolayer CrCl_3 system, we turn to the realistic charge doping by adsorption in the following subsections. Here we choose Li (F) atoms as electron (hole) donors in our studies and study the effect of adsorption on the electronic structure and magnetic properties of monolayer CrCl_3 .

3.2. Magnetic properties of Li-adsorbed CrCl_3

First of all, to obtain the most energetically favorable adsorption configuration, we have considered three adsorption sites labeled as (s1) hollow (at the hollow of honeycomb network of Cr atoms), (s2) Cr-top (at the top of Cr atom), and (s3) Cl-top (at the top of Cl atom), as shown in Fig. 1(a) and (b). After fully relaxation, as listed in Table 1, the hollow site configuration has the lowest energy (~ 3.642 eV), and the Cl-top configuration is the most unfavored (~ 3.030 eV). Based on the current results, the hollow site configuration is considered as the most stable configuration. The impact of Li coverage on the electronic and magnetic properties of monolayer CrCl_3 is investigated. When Li coverage changes from 0% to 12.5%, as shown in Fig. 3(a), the magnetism is almost linearly enhanced as the Li coverage increases, along with the local spin moment of the Cr. It is clear that the total magnetic moment is mainly contributed by Cr atoms, with $3.5 \mu_B$ for each Cr at the 12.5% Li-coverage, much higher than that of the pristine one ($\sim 3.0 \mu_B$). Besides, the total energy difference between the stable AFM and FM configuration is shown in Fig. 3(b). It is found that the energy difference ($E_{\text{AFM}} - E_{\text{FM}}$) is becoming much larger as the Li-coverage increases. As Li-coverage increases from 0% to 12.5%, it changes from 27 meV to 181 meV. The 0% coverage means the pristine CrCl_3 monolayer. Our results indicate the most stable state is always FM, and the FM coupling in CrCl_3 -Li is enhanced by Li adsorption. These results further confirm that the magnetism can be enhanced after Li adsorption.

As we mentioned above, the magnetism is almost linearly enhanced as the Li coverage increases. In our opinion, the enhancement of the magnetism could be understood as below. In every octahedral coordination, the Cr-d orbitals split into triply degenerated t_{2g} (d_{xy} , d_{yz} , and d_{zx}) and doubly degenerated e_g (d_{z^2} and $d_{x^2-y^2}$), the former with lower energy. In pristine CrCl_3 , each coordinated Cr has a $3d^3$ configuration, and the three t_{2g} orbitals are singly occupied with parallel spin. Whereas in CrCl_3 -Li, due to an electron doping effect, besides three electrons occupy the three t_{2g} orbitals, the other electrons occupy the higher e_g orbital according to Hund's rules, thus resulting in larger magnetic moment. Also, the charge density and PDOS of one Li-adsorbed CrCl_3 are plotted in Fig. 4. As expected, Li atoms provide electrons to CrCl_3 monolayer, as shown in Fig. 4(a) and (b). These changes in local structure will influence the electronic structure. As illustrated in Fig. 4(c), the Li-s and Cl-p orbitals are coupled, indirectly altering the hybridization between Cl-p and Cr-d orbitals. Comparing Fig. 1(c) with 4(c), we find that the Fermi level is pushed up and cross the spin-polarized conduction bands in CrCl_3 -Li. In current cases, the Li-Cl charge transfer indirectly results in Li atoms donating partial electrons to Cr atoms, which brings about an electron doping effect. These results indicate that charge transfer due to Li adsorption can significantly affect the electronic properties of CrCl_3 . These results further confirm that the magnetism can be enhanced after Li adsorption.

Finally, we investigate how MAE changes with Li coverage. When Li

Table 1

Adsorption energy (E_{ads}) of Li- and F-adsorbed $2 \times 2 \times 1$ CrCl_3 monolayer.

Configuration		Hollow (s1)	Cr-top (s2)	Cl-top (s3)
E_{ads} (eV)	Li	-3.642	-3.215	-3.030
	F	-1.164	-1.110	-1.110

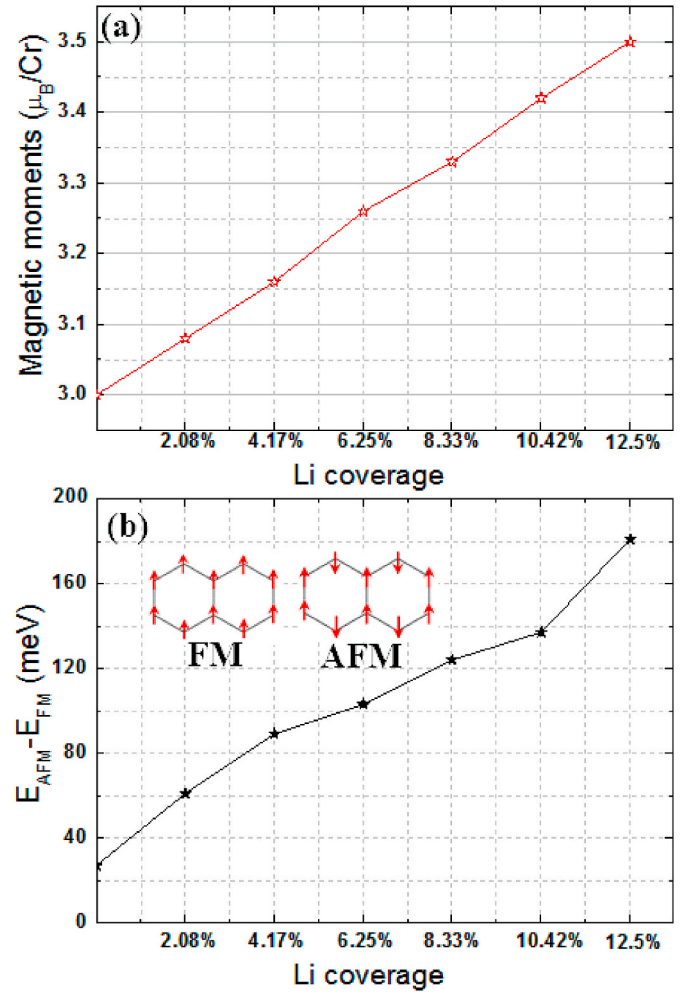


Fig. 3. (a) The local magnetic moment of Cr and (b) total energy differences between AFM and FM states in the Li-adsorbed $3 \times 2 \times 1$ CrCl_3 monolayer. Two magnetic configurations, named ferromagnetic (FM) state and antiferromagnetic (AFM) states, are shown.

is adsorbed onto CrCl_3 monolayer, the MAE values change from positive to negative at certain coverage (between 4.17% and 6.25%) and increase negatively and monotonically with increasing Li adsorption coverage, as shown in Fig. 5. For example, the MAE changes from 0.789 meV/Cr to -2.303 meV/Cr while Li adsorption coverage increases from 0% to 12.5%. These results clearly show that we can effectively switch the easy magnetization orientation from out-of-plane to the in-plane direction by adsorbing Li atoms. We attribute the change of MAE to the electron doping effect induced by charge transfer between Li and Cr atoms.

3.3. Magnetic properties of F-adsorbed CrCl_3

The adsorption of F atoms on monolayer CrCl_3 is considered to simulate the practical hole doping. Since the electronegativity of F is stronger than Cl, direct charge transfer may occur between F and Cr, thus leading to hole doping. To confirm the most stable adsorption configuration, we also considered three initial adsorption configuration as mentioned above. It is found that F atoms always relax far away from the initial high symmetrical sites, indicating strong re-construction in our DFT calculations. To qualitatively reveal the effect of the hole doping and compare with CrCl_3 -Li, the MAE of CrCl_3 -F was also calculated with F atoms fixed at the hollow site as in CrCl_3 -F. In contrast to that of CrCl_3 -Li, F adsorption significantly reduce the magnetism. As

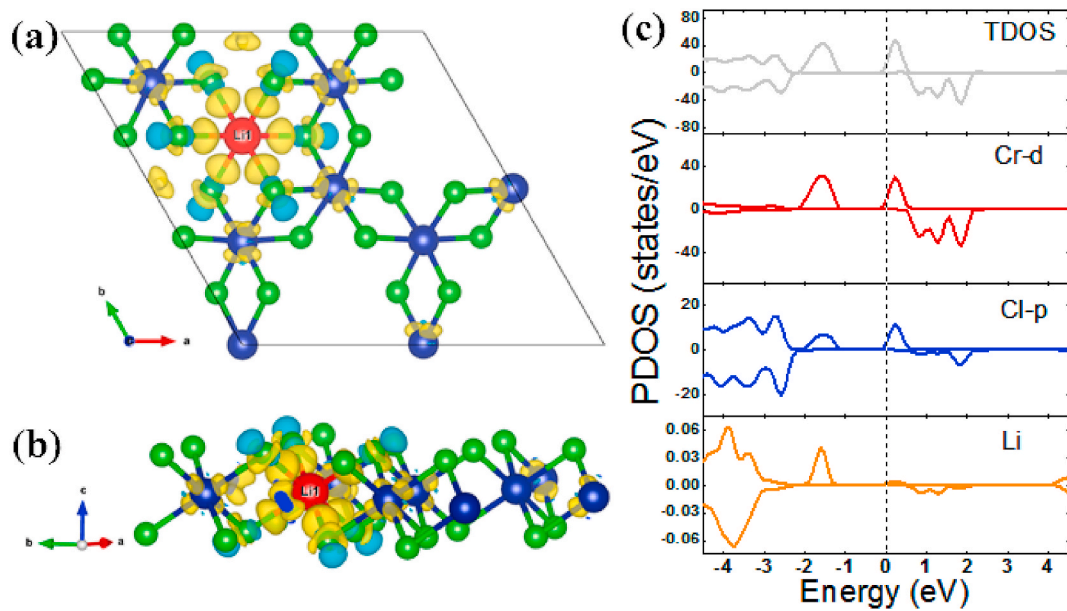


Fig. 4. (a) Top and (b) side views of spin density distribution; (c) PDOS in one-Li-adsorbed CrCl₃ monolayer. Yellow and cyan isosurfaces represent positive and negative spin densities ($0.002 \text{ e}/\text{\AA}^3$), respectively. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

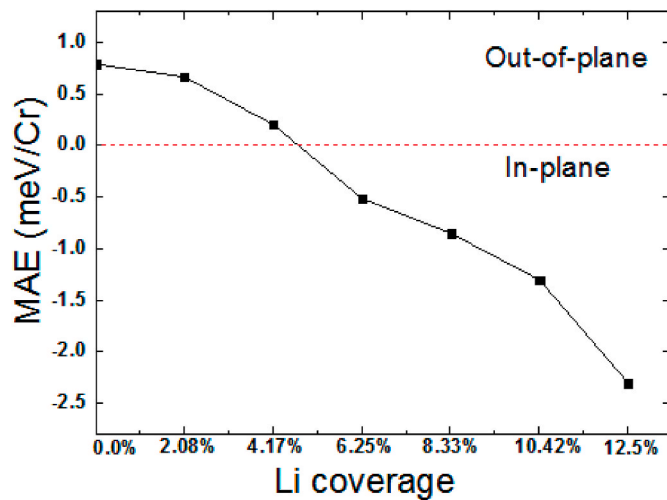


Fig. 5. Magnetic anisotropy energy (MAE) of the $3 \times 2 \times 1$ Li-CrCl₃ monolayer with different Li adsorption coverage.

shown in Fig. 6(a), the magnetism is almost linearly reduced as the F coverage increases, and the total magnetic moment reduces to $2.5 \mu_B/\text{Cr}$ at the 12.5% F-coverage. Moreover, as shown in Fig. 6(b), the F adatoms enhance the MAE, but not able to alter the magnetic orientation. For example, the MAE of CrCl₃-F is as large as 2.906 meV/Cr and 1.512 meV/Cr when F coverage is 6.25% and 12.5% respectively. Such a difference can be well expected from the electronegativity of Li and F atom. These results show that electrostatic doping induced by surface adsorption could tune the MAE of monolayer CrCl₃. Especially, electron doping can drastically tune the spin orientations. Therefore, the strong coupling between charge transfer and MAE provides an effective way to control spin orientation, which provides a very promising means for the application of spin electronics devices in the future.

4. Conclusions

In summary, the electronic and magnetic properties of monolayer

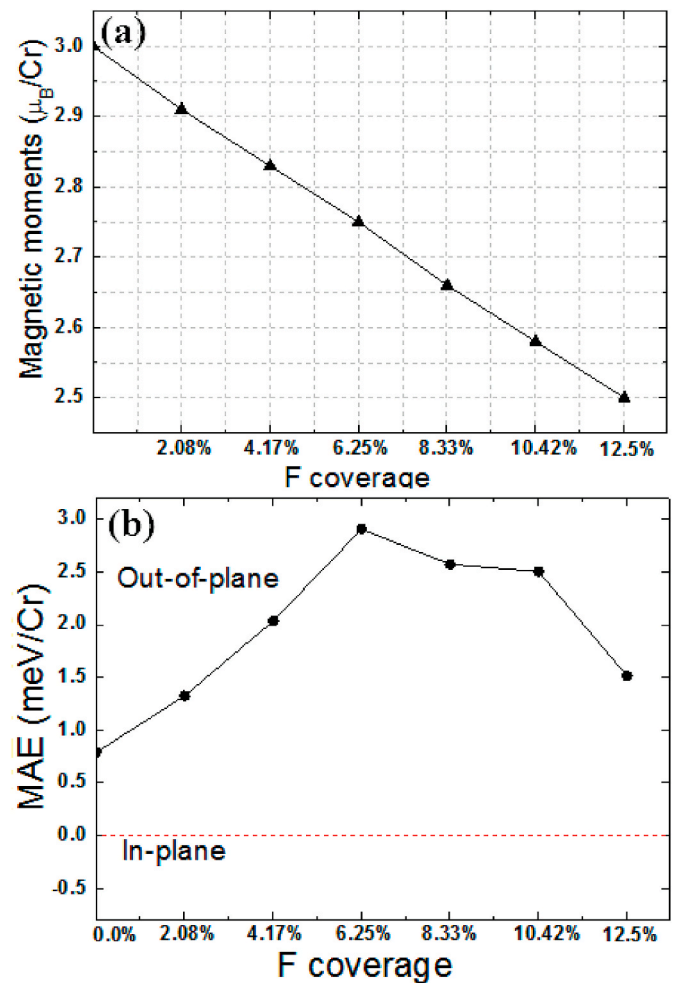


Fig. 6. (a) The local magnetic moment of Cr and (b) MAE of the $3 \times 2 \times 1$ F-CrCl₃ monolayer with different F adsorption coverage.

CrCl₃ could be tuned by the surface adsorption. Our results show that the magnetic moments on Cr can be largely enhanced by Li adsorption. Interestingly, the CrCl₃-Li exhibits robust magnetic anisotropy (-2.303 meV/Cr) with an in-plane easy magnetization axis, almost 3 times compared with that of the pristine CrCl₃ (0.789 meV/Cr) with an out-of-plane easy axis. Moreover, the value of MAE increases linearly with the increment of the Li adsorption coverage. On the contrary, the F adsorption can reduce the saturation magnetization, but preserving the easy magnetization axis. The robust MAE together with the enhanced saturation magnetization of monolayer CrCl₃ induced by Li adsorption may provide promising applications in 2D spintronic devices.

CRediT authorship contribution statement

M. Luo: Methodology, Writing - review & editing. **Y.D. Li:** Software, Validation, Data curation, Writing - review & editing. **K.J. Wang:** Visualization, Investigation, Writing - review & editing. **Y.H. Shen:** Conceptualization, Supervision, Writing - review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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